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Optimal Diagnostic Questionnaires

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This paper develops dynamic-programming and other shortest-route algorithms for optimizing sequentially patterned questioning procedures, called diagnostic questionnaires. It uses graph-theoretic methods and solves certain combinatorial problems associated with the Catalan numbers. The paper obtains general properties of optimal questionnaires, including an ordering relation, a convexity result, and the determination of essentially complete classes of questionnaires. It explores a logarithmic charging scheme in detail and obtains a minimax result. Finally, the paper examines the relation with Huffman coding of information theory, and a simple method of Shannon efficiency.

THIS PAPER PROVIDES a general framework for the optimization of sequentially patterned questioning procedures, called by PICARD^[27] and DUNCAN^[11,12] (*diagnostic*) *questionnaires*. Questionnaires have applications ranging over medical diagnosis,^[27] taxonomic keys,^[21,24] group tests,^[30] weighing designs,^[2] "twenty questions" game strategies,^[8] trouble-shooting routines,^[6] computer programming,^[13,17] search theory,^[31,32] and sort routines.^[5,14] Each questionnaire considered here has a tree representation, as illustrated in Figs. 1 and 2.

A charge structure for the use of questions as outlined in Section 1 poses the basic optimization problem. Section 2 gives necessary conditions for optimality and establishes useful, essentially complete class results. Combinatorial problems arise naturally, generally associated with the historic Catalan numbers (*see, e.g.,* ORDMAN^[23]); results concerning these problems are given in Section 3. In Section 4 we demonstrate that optimization can be formulated as an extension of the shortest-route problem (*see, e.g.,* BELLMAN, ET AL.^[3]), and hence can be solved by dynamic programming or other shortest-route algorithms. A dynamic-programming algorithm is given in Section 5, as well as a generalized permanent label-setting algorithm due to NEMHAUSER.^[22] A specific charging scheme, logarithmic charging, is discussed in Section 6, establishing a link between questionnaires and information theory. This link is based on the fact that a questionnaire can be viewed as a code constructed from several alphabets, each with a different number of characters. This generalizes the usual noiseless coding theory (*see, e.g.,* ASH^[1]).

1. BASIC NOTIONS

A COMMON ELEMENT in each application is the existence of a *true* state θ to be found among a finite set of possible states $\Theta = \{\theta_1, \dots, \theta_m\}$. Each state θ_i occurs with known probability $p_i > 0$, where $p_1 + \dots + p_m = 1$. The search for θ is carried out by sequential choice of questions from the set $\mathcal{Q} = \{q_1, \dots, q_r\}$ of available questions.

Each question is here taken to be a nontrivial partition of some subset of θ (for a more general formulation, see Duncan,^[12] if the restriction “nontrivial” is dropped, a quasiquestionnaire results—see Picard^[28]). In most practical applications not all partitions will be available in $\mathcal{Q}$ (see SOBEL^[30] on group testing and HADIAN^[14] on deterministic ranking). Use of the question q_j entails a known charge c_j , which will usually be a positive constant. Questions will be assumed to elicit (upon payment of the appropriate charge) a truthful response as to whether θ is

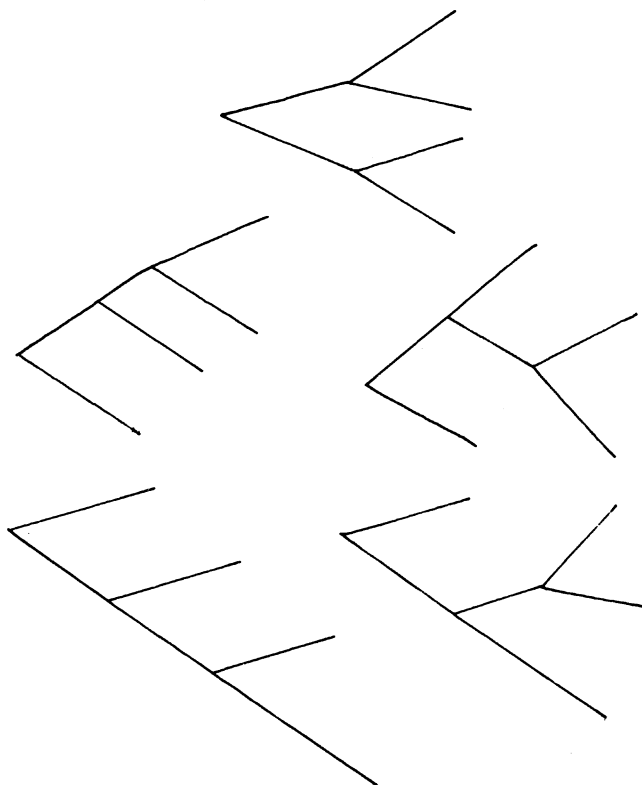


Fig. 1. Binary questionnaires with $m = 4$.

located in the partition. A questionnaire is a collection of questions that then surely locate θ by resolving Θ into the singleton sets $\{\theta_1\}, \dots, \{\theta_m\}$. An optimal questionnaire is defined as one with minimum average charge. Some results about the structure of optimal questionnaires are given by Duncan,^[11,12] PETOLLA,^[26] and PARKHOMENKO.^[25] The special homogeneous case in which each question is binary, i.e., only dichotomous partitions are available in $\mathcal{Q}$, is studied by GAREY^[13] and HU AND TAN.^[17]

It is generally fruitful to employ graph-theoretic representations. The questionnaires we shall consider are rooted, directed trees, i.e., arborescences, with m terminal nodes. Figures 1 and 2 illustrate such representations.

2. GENERAL PROPERTIES OF OPTIMAL QUESTIONNAIRES

SUPPOSE ARBITRARY FIXED charges c_j are made for each question q_j as it is asked. Then Duncan^[12] has shown that an optimal questionnaire can be found in a class restricted by three conditions, where the states are relabelled in decreasing order of probability so that $p_1 \geq p_2 \geq \dots \geq p_m$. Let ϕ_i be the total charge imposed when $\theta = \theta_i$, i.e., the sum of the charges c_j for the questions q_j required to reach θ_i .

THEOREM 2.1. *There exists an essentially complete class of questionnaires satisfying the following three conditions: (1) The total charges are ordered so that $\phi_i \leq \phi_{i+1}$,*

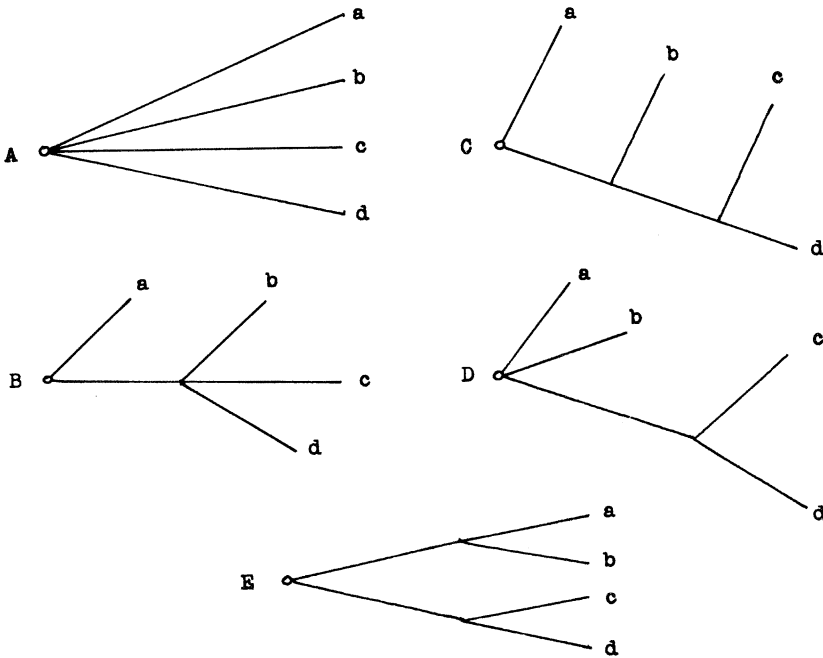


Fig. 2. Heterogeneous order-constrained questionnaires with $m = 4$.

$i = 1, \dots, n$. (2) Each question is a partition in index-contiguous blocks. (3) The number of elements in these blocks is a nondecreasing function of their induced index.

The proof of this theorem is based on showing no decrease in average charge when nonconforming states are interchanged. The fruitful notion of an essentially complete class of questionnaires has been discussed by Picard (reference 29, problem 8).

In investigating whether a proposed procedure does in fact give optimal results, it is helpful to introduce some notation and obtain a convexity result. If Q is a questionnaire and p a probability vector, denote by $E_p C(Q)$ the average charge for Q at p .

DEFINITION 2.1. Let A be a subset of the probability simplex, $p_0 \in A$, and Q_0 have minimum charge at p_0 . Then A is called a region of minimum average charge if $A = \{p : E_p C(Q_0) \leq E_p C(Q) \text{ for any } Q\}$.

THEOREM 2.2. *Regions of minimum average charge are convex.*

Proof. Use the method of BLACKWELL AND GIRSHICK (reference 4; Theorem 6.3.1., p. 150). A further restriction on the class of questionnaires that need be examined is given by the next result.

THEOREM 2.3. *The set of questionnaires $\mathfrak{B}$ whose average charge depends on p only through $p_1, p_2, \dots, p_{m-2}$ is an essentially complete class.*

Proof. Choose $Q \notin \mathfrak{B}$. Then there exist p and p' , differing only in the last two coordinates, such that $p_{m-1}\phi_{m-1} + p_m\phi_m \neq p'_{m-1}\phi_{m-1} + p'_m\phi_m$. Using family-tree terminology, we note that $\{\theta_{m-1}\}$ and $\{\theta_m\}$ are not offspring of the same node (requiring $\phi_{m-1} = \phi_m$), since $p_{m-1} + p_m = p'_{m-1} + p'_m$. Therefore, there exists a state $\theta_i \in \{\theta_1, \dots, \theta_{m-2}\}$ that is a sibling of θ_m , since no node can be an only child. We now show that $\tilde{Q}$, which interchanges the role of θ_i and θ_{m-1} in Q , is both in $\mathfrak{B}$ and at least as good as Q . First, $E_p C(\tilde{Q})$ depends on p_{m-1} and p_m only through $p_{m-1} + p_m = 1 - (p_1 + \dots + p_{m-2})$, and hence is in $\mathfrak{B}$. Second, the statement $E_p C(\tilde{Q}) \leq E_p C(Q)$ is equivalent to $p_i\phi_{m-1} + (p_{m-1} + p_m)\phi_m \leq p_{m-1}\phi_{m-1} + (p_i + p_m)\phi_m$, which in turn is equivalent to $\phi_m \geq \phi_{m-1}$. This last statement is affirmed for an essentially complete class of questionnaires by Theorem 2.1.

TABLE I
THE NUMBER OF NONTRIVIAL UNORDERED PARTITIONS OF THE INTEGER n

n	4	5	10	25	50	100
$p(n)-1$	4	6	41	1957	204,225	190,569,291

3. COMBINATORIAL ASPECTS OF QUESTIONNAIRES

THE SET-PARTITION nature of a questionnaire suggests that combinatorial problems about questionnaires have number-theoretic partition analogs. In fact, it is easily seen that, for each question on a set with, say, n elements, there are just as many (set-theoretic) partitions that satisfy the last two conditions of Theorem 2.1 as there are (number-theoretic) nontrivial unordered partitions of the integer n . The combinatorial problem of finding the number of such partitions is discussed, for example, in HALL^[15] and some typical values are given in Table I.

The asymptotic expression for $p(n)$,

$$(1/4n\sqrt{3})\exp[\pi(2n/3)^{1/2}],$$

is the first term of a series representation proposed by HARDY AND RAMANUJAN.^[16] LEHMER^[19] showed divergence of the series and reference 20 gave systematic methods for generating the partitions.

Related combinatorial problems arise when we begin to count questionnaires. We shall consider three basic problems; in each a recursion relation is easily developed.

The first problem is to count the total number of trees that represent binary questionnaires on a state space with m elements. The recursion relation that must be satisfied is $u_m = \sum u_j u_k, j+k=m$, with the initial condition $u_1=1$. This gives rise to the familiar Catalan numbers (see, e.g., Ordman^[23]). Hall [reference 15,

p. 26] derives the well known result $u_m = (2m-2)!/(m-1)!m!$. We note that $u_4 = 5$ and display the five binary questionnaires in Fig. 1.

The second problem is to count the total number v_m of trees that represent questionnaires, with questions of arbitrary resolution, on a state space with m elements. The recursion relation is $v_m = \sum v_{j_1} \cdots v_{j_k}, j_1 + \cdots + j_k = m$, with the initial condition $v_1 = 1$.

The sequence of numbers $\{v_m\}$ can be shown to have the generating function $f(s) = \sum v_m s^m$ satisfying $f^2(s) + f^3(s) + \cdots = f(s) - s$, or, equivalently, $f^2(s)/[1 - f(s)] = f(s) - s$. This yields $f(s) = (\frac{1}{4})[s + 1 - \sqrt{s^2 - 6s + 1}]$, or expanding in a Taylor series,

$$f(s) = (\frac{1}{4}) \left[s + 1 - \sum_{x=0}^{\infty} \binom{\frac{1}{2}}{x} (s^2 - 6s)^x \right].$$

The third combinatorial problem is to count the number w_m of questionnaires in the essentially complete class that satisfy the last two conditions of Theorem 2.1. As in the previous problems, a recursion relation is available for w_m . Note that each questionnaire has an initial question of some resolution k , $2 \leq k \leq m$. There

TABLE II
THE NUMBER OF QUESTIONNAIRES w_m IN AN ESSENTIALLY COMPLETE CLASS

m	w_m	m	w_m	m	w_m
1	1	8	277	15	776494
2	1	9	806	16	2545777
3	2	10	2500	17	8223352
4	5	11	7578	18	27221690
5	12	12	24198	19	88984144
6	34	13	75370	20	296856400
7	92	14	243800	21	979829772

are then k subquestionnaires resolving k state spaces $\theta_1, \cdots, \theta_k$ with, say, $j_1, \cdots, j_k$ elements, respectively, where $j_1 + \cdots + j_k = m$. Theorem 2.1 then requires that w_m satisfy the recursion relation $w_m = \sum w_{j_1} \cdots w_{j_k}$, where the sum is taken over all k such that $2 \leq k \leq m$ and over all j_i such that $j_1 + \cdots + j_k = m$ and $1 \leq j_1 \leq \cdots \leq j_k$. The initial condition $w_1 = 1$ must be satisfied. Thus, the third problem is identical to the second with the exception of the order constraint imposed on the indices here.

An algorithm based on this recursion relation was programmed for the Burroughs 6700 computer to obtain some of the 'small' values of w_m . Fifteen seconds of computing time produced the numerical values of w_1 through w_{21} , the last value indicating that a state space with 21 elements has just short of one billion questionnaires in the essentially complete class defined over it. The exact values are given in Table II. Noting that $w_4 = 5$, we display the five questionnaires in Fig. 2.

4. THE SHORTEST-ROUTE FORMULATION

THIS SECTION SHOWS that questionnaire optimization is an extension of the shortest-route problem (see, e.g., Bellman et al.^[3]). Hence, an optimal questionnaire can be found using dynamic-programming methods or other shortest-route algorithms.

First, note that each question effects a refinement of the partitions of Θ . The relation of refinement by questions defines a partial ordering on the subsets of Θ . Conditions (2) and (3) of Theorem 2.1 determine that only certain partitions need be considered. We can associate with each such set partition a number-theoretic partition of the integer $m = |\Theta|$. This association is bijective, i.e., one-to-one and onto. A graph-theoretic representation is then possible by establishing a node for each required partition and a directed edge from each node to each finer node. Call this graph G . Note that a question or set of questions at the same level then corresponds to a directed edge in G , with one's state of knowledge summarized, before and after asking, by the two end nodes. Each path from the node for the trivial partition to the node for the total partition into singleton sets defines a questionnaire.

These ideas are best illustrated by an example. Consider the case $\Theta = \{a, b, c, d\}$. The various correspondences are given in Table III, while the graph G is shown in Fig. 3. Note that the arrowheads in Fig. 3 function as probabilistic toll gates in the assessment of charges. Thus the arrowhead entering node 5 from node 3 assesses a charge of 0 with probability p_1 and the specified charge for resolving $\{b, c, d\}$ into $\{b\}, \{c\}, \{d\}$ with probability $1 - p_1$. Corresponding to each path from node 1 to node 5 in G there is an admissible questionnaire. These are shown in Fig. 2.

TABLE III
THE CORRESPONDENCES FOR THE EXAMPLE

Set partition	Partition of the integer 4	Node number
$\{a, b, c, d\}$	4	1
$\{a, b\}, \{c, d\}$	2+2	2
$\{a\}, \{b, c, d\}$	1+3	3
$\{a\}, \{b\}, \{c, d\}$	1+1+2	4
$\{a\}, \{b\}, \{c\}, \{d\}$	1+1+1+1	5

Given a probability vector p on Θ and a particular charging scheme, there is an average charge for transversing the edge joining node i and node j in the graph G , our goal being to find a minimum-average-charge questionnaire. This then completes a shortest-route formulation of the problem. (We note that, if all questions are not available, the effect is to remove edges from the graph G . Unfortunately, for large state spaces this usually makes the problem more difficult rather than easier and it must be studied on an *ad hoc* basis.)

The dynamic-programming algorithm is then based on the functional equation

$$f_i = \min_{j>i} [t_{ij} + f_j], \quad (1)$$

where f_i is the minimum average charge for moving from level i (node i in G) to complete resolution of θ [node $p(m)$ in G]. The standard iterative algorithm based on (1) is discussed in the next section, as well as an alternative shortest-route algorithm.

5. OPTIMIZATION METHODS

THE SIMPLE METHOD of successive comparison is easily ruled out as an optimization algorithm, since a state space with m elements requires $w_m - 1$ (w_m was discussed in

Section 4) binary comparisons, even restricting the class of questionnaires according to the last two conditions of Theorem 2.1. Better is a dynamic-programming algorithm based on the recursive relation

$$f^{(k)}(i) = \min_{i < j \leq n} [t_{ij} + f^{(k-1)}(j)],$$

where n is the number of nodes in the graph G . The number n is just the number of partitions of the integer m and hence is $p(m)$, a quantity discussed in Section 4. It is shown, for example in Bellman et al.,^[3] that the recursion algorithm yields con-

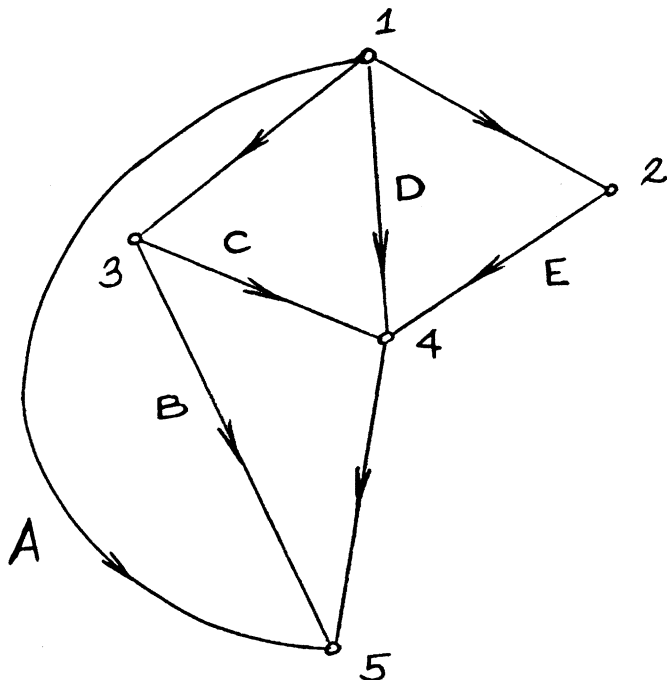


Fig. 3. The shortest-route formulation of the example.

vergence to an optimum in no more than $n = p(m)$ iterations. This represents a substantial improvement over the successive-comparison algorithm and is certainly a workable method for $m = 20$, say. However, when m is about 50, the amount of computation required is excessive. A more efficient shortest-route algorithm has been proposed by DIJKSTRA.^[9]

Dijkstra's algorithm is reviewed and compared to others by DREYFUS.^[10] We shall consider a generalization of this algorithm due to Nemhauser.^[22] The algorithm is *permanent label setting* in that the set of nodes $N = \{1, \dots, n\}$ is partitioned into P , the set of permanently labelled nodes and T , the set of temporarily labelled nodes. At each iteration additional nodes are transferred from T into P . Input to the algorithm is a real-valued function h with domain $N - \{1\}$ satisfying the property $h(i) \leq h(j) + t_{ij}$ for all $i, j \in N - \{1\}$.

Step 0. Initialize $P = \{1\}$, $T = \{2, \dots, n\}$, $l(1) = 0$, $l(j) = t_{ij}$, $j \in T$.

Step 1. Let $J = \{j^*: l(j^*) + h(j^*) = \min_{j \in T} [l(j) + h(j)]\}$.

Step 2. If $n \in J$ terminates, $l(n) = \text{average charge for the optimal questionnaire}$. [To obtain the optimal path, work backward from node n , choosing nodes i and j in P such that $|l(i) - l(j)| = t_{ij}$.] If $n \notin J$, let $T = T - J$ and $P = P \cup J$.

Step 3. For each $j \in T$, let $l(j) = \min_{j^* \in J} [l(j^*) + t_{j^*j}]$. Return to Step 1.

We note the important fact that, as a consequence of their definition, $h(i) \leq f(i)$, and hence the function $h(i)$ can be chosen to be a lower bound on the average charge for moving from node i to node n . As pointed out by Nemhauser, the choice of a nonzero $h(i)$ can decrease the number of required computations. The partial ordering of nodes in N can be used to reduce the number of computations required. It is only necessary to label permanently the nodes that are not less than some node already permanently labelled.

6. THE LOGARITHMIC CHARGING SCHEME

VARIOUS INTERESTING CONNECTIONS can be made between questionnaire theory and information theory by specifying a charge of $\log d$ for each question of resolution d . (A convenient convention that entails no loss of generality is to use logarithms to the base two.) Simple illustrations of this connection are given in Ash^[1] and it has been explored more fully by Picard.^[27] Sobel^[30] has used this connection profitably in the group-testing context. The basic quantity of importance is the Shannon entropy, defined for each probability vector p by $H(p) = -\sum p_i \log p_i$. We begin with three basic facts proved in this generality by Duncan.^[11,12]

THEOREM 6.1 (noiseless coding). *Any questionnaire has an average charge bounded below by the Shannon entropy.*

THEOREM 6.2 (Shannon efficiency). *The Shannon lower bound is attained by a questionnaire if and only if each question is Shannon efficient.* (A resolution d question is called *Shannon efficient* if each of its d sets has equal probability.)

THEOREM 6.3 (upper bound). *There exists a questionnaire with average charge less than the Shannon entropy plus one. This bound is sharp.*

A minimax result is available as the next result.

THEOREM 6.4 (minimax). *The questionnaire Q^* consisting of an initial resolution m question is minimax, $p^* = (1/m, \dots, 1/m)$ is a least favorable distribution, and $\log m$ is the value of the game.*

Proof. Note that $H(p^*) = E_{p^*} C(Q^*) = \log m$. Theorem 6.1 then shows that Q^* is a Bayes rule. It is an equalizer rule with constant risk $\log m$.

6.1. Huffman Coding

Suppose a questionnaire is homogeneous in that every question is of the same fixed resolution d . Then average-charge minimization is equivalent to average-length minimization, and the well known HUFFMAN coding scheme^[18] can be used to obtain an optimal questionnaire. This optimization procedure has been gen-

eralized by Picard^[27] to the class of heterogeneous questionnaires in which the number of questions of each resolution to be used is fixed a priori. Different costs are introduced by Petolla.^[26] It is not clear that the Huffman procedure admits a generalization to optimization in the broader class of arbitrary-resolution heterogeneous questionnaires.

6.2. The Method of Shannon Efficiency

The Huffman coding scheme is essentially a backward optimization procedure. A very simple forward procedure would attempt to use Theorem 6.2 and construct Shannon-efficient questions to the farthest possible stage. Picard^[27] has given a complicated example to show that this procedure can fail in the binary case. A much simpler example of failure is available for $m=4$ with heterogeneous questionnaires. The regions suggested by this procedure are displayed in Duncan;^[11] they are not convex, and hence the procedure fails by Theorem 2.2. Nevertheless, an attempt in the direction of Shannon efficiency should be made—as is carried out in the following result.

THEOREM 6.5. *At any stage of questioning, the optimal question (partition of the conditional state space) must satisfy*

$$\log d - 1 < H(\rho_1, \dots, \rho_d) \leq \log d,$$

where a resolution d question is asked and the offspring have probabilities $\rho_1, \dots, \rho_d$. This gives a necessary condition for optimality.

To prove this theorem it is helpful to have the following lemma.

LEMMA 6.1. *If $K(p)$ denotes the minimum average charge at p and the optimal questionnaire calls for a resolution d question, then*

$$K(p) = \log d + \sum \rho_j K(\rho^{(j)}),$$

where $\rho^{(j)}$ is the probability vector of the elements of the j th offspring.

Proof. Use $E_p C(Q) = \log d + \sum \rho_j E_{\rho^{(j)}} C(Q_j)$ and the definition of $K(p)$.

Proof of Theorem 6.5. It is sufficient to consider the first stage of questioning. By Theorem 6.3, $K(p) < H(p) + 1$. We now apply Lemma 6.1 and the well known defining property of Shannon entropy, the generalized grouping axiom, to obtain

$$\log d + \sum \rho_j K(\rho^{(j)}) < H(\rho_1, \dots, \rho_d) + \sum \rho_j H(\rho^{(j)}) + 1.$$

This implies

$$\sum \rho_j [K(\rho^{(j)}) - H(\rho^{(j)})] < H(\rho_1, \dots, \rho_d) - \log d + 1. \quad (2)$$

But the left-hand side of (2) is nonnegative by Theorem 6.1, and hence

$$0 < H(\rho_1, \dots, \rho_d) - \log d + 1.$$

This, together with the fact that $H(\rho_1, \dots, \rho_d) \leq \log d$ gives the theorem.

The theoretical results presented in this section give some plausible initial guesses for use in the dynamic-programming iterative solution discussed in Section 5. We know that

$$H(\rho^{(i)}) \leq f_i \leq \min[\log m_i, H(\rho^{(i)}) + 1],$$

where, at stage i , θ has been resolved into m_i sets with probability vector $\rho^{(i)}$. Thus, one might choose the midpoint of these two bounds as $f_i^{(0)}$.

In the permanent-label-setting algorithm of Nemhauser, a lower bound $h(i)$ is available from Theorem 6.1 as the Shannon entropy of the set partition indicated by node i .

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